

Bexiprost 0.005% w/v Eye Drops

Latanoprost 0.05mg

Description

Colourless or pale yellow, clear solution.

Pharmacodynamics

Pharmacotherapeutic group: Antiglaucoma preparations and miotics, prostaglandin analogues ATC code: S01EE01

The active substance latanoprost, a prostaglandin $F_{2\alpha}$ analogue, is a selective prostanoid FP receptor agonist which reduces the intraocular pressure by increasing the outflow of aqueous humour. Reduction of the intraocular pressure in man starts about three to four hours after administration and maximum effect is reached after eight to twelve hours. Pressure reduction is maintained for at least 24 hours.

The main mechanism of action is increased uveoscleral outflow, although some increase in outflow facility (decrease in outflow resistance) has been reported in man.

Latanoprost is effective as monotherapy. In addition, latanoprost is effective in combination with beta-adrenergic antagonists (timolol). The effect of latanoprost is additive in combination with adrenergic agonists (dipivalyl epinephrine), oral carbonic anhydrase inhibitors (acetazolamide) and at least partly additive with cholinergic agonists (pilocarpine).

Latanoprost has no significant effect on the production of aqueous humour. Latanoprost has not been found to have any effect on the blood-aqueous barrier.

Latanoprost has no or negligible effects on the intraocular blood circulation. However, mild to moderate conjunctival or episcleral hyperaemia may occur during topical treatment.

Latanoprost has not induced fluorescein leakage in the posterior segment of pseudophakic human eyes during short-term treatment.

Latanoprost in clinical doses has not been found to have any significant pharmacological effects on the cardiovascular or respiratory system.

Pharmacokinetics

Latanoprost (mw 432.58) is an isopropyl ester prodrug which per se is inactive, but after hydrolysis to the acid of latanoprost becomes biologically active.

The prodrug is well absorbed through the cornea and all drug that enters the aqueous humour is hydrolysed during the passage through the cornea.

The peak concentration in the aqueous humour is reached about two hours after topical administration. After topical application, latanoprost is distributed primarily in the anterior segment, the conjunctivae and the eyelids. Only minute quantities of the drug reach the posterior segment.

There is practically no metabolism of the acid of latanoprost in the eye. The main metabolism occurs in the liver. The half life in plasma is 17 minutes in man. The main metabolites, the 1,2-dinor and 1,2,3,4-tetranor metabolites, exert no or only weak biological activity in animal studies and are excreted primarily in the urine.

Paediatric population

Latanoprost acid systemic exposure was approximately 2-fold higher in 3 to < 12 year olds and 6-fold higher in children < 3 years old compared with adults, but a wide safety margin for systemic adverse effects was maintained. Median time to reach peak plasma concentration was 5 minutes post-dose across all age groups. The median plasma elimination half-life was short (< 20 minutes), similar for paediatric and adult, and resulted in no accumulation of latanoprost acid in the systemic circulation under steady-state conditions.

Indications

- Reduction of elevated intraocular pressure (IOP) in patients with open-angle glaucoma, chronic angle closure glaucoma, and ocular hypertension.
- Reduction of elevated intraocular pressure in paediatric patients with elevated intraocular pressure and paediatric glaucoma

Recommended Dosage

Use in adults (including the elderly):

One drop in the affected eye(s) once daily. Optimal effect is obtained if latanoprost is administered in the evening.

The dosage of latanoprost should not exceed once daily since it has been shown that more frequent administration decreases the intraocular pressure lowering effect.

If one dose is missed, treatment should continue with the next dose as normal.

Latanoprost may be used concomitantly with other classes of topical ophthalmic drug products to lower IOP. If more than one topical ophthalmic drug is being used, the drugs should be administered at least five minutes apart.

Contact lenses should be removed before instillation of the eye drops and may be reinserted after fifteen minutes (because this product contains benzalkonium chloride)

Paediatric Population

Latanoprost eye drops may be used in paediatric patients at the same posology as in adults. No data are available for preterm infants (less than 36 weeks gestational age). Data in the age group < 1 year are limited.

Method of administration

As with any eye drops, to reduce possible systemic absorption, it is recommended that the lachrymal sac be compressed at the medial canthus (punctal occlusion) for one minute. This should be performed immediately following the instillation of each drop.

Contact lenses should be removed before instillation of the eye drops and may be reinserted after 15 minutes.

If more than one topical ophthalmic medicinal product is being used, the medicinal products should be administered at least five minutes apart.

Route of Administration: Ophthalmic

Contraindications

Known hypersensitivity to latanoprost, benzalkonium chloride or any other ingredients in this product.

Warning and Precautions

Bexiprost may gradually change eye colour by increasing the amount of brown pigment in the iris. Before treatment is instituted, patients should be informed of the possibility of a permanent change in eye colour. Unilateral treatment can result in permanent heterochromia.

This change in eye colour has predominantly been seen in patients with mixed coloured irides, i.e. blue-brown, grey-brown, yellow-brown and green-brown. The onset of the change is usually within the first 8 months of treatment, rarely during the second or third year, and has not been seen after the fourth year of treatment. The rate of progression of iris pigmentation decreases with time and is stable for five years. The incidence in patients with mixed colour irides ranged from 7 to 85%, with yellow-brown irides having the highest incidence. The colour change is due to increased melanin content in the stromal melanocytes of the iris and not to an increase in number of melanocytes. Typically, the brown pigmentation around the pupil spreads concentrically towards the periphery in affected eyes, but the entire iris or parts of it may become more brownish. No further increase in brown iris pigment has been observed after discontinuation of treatment. It has not been associated with any symptom or pathological changes.

Neither naevi nor freckles of the iris have been affected by treatment. Accumulation of pigment in the trabecular meshwork or elsewhere in the anterior chamber has not been observed. Increased iris pigmentation has not been shown to have any negative clinical sequelae and Bexiprost can be continued if iris pigmentation ensues. However, patients should be monitored regularly and if the clinical situation warrants, Bexiprost treatment may be discontinued.

There is limited experience of Bexiprost in chronic angle closure glaucoma, open angle glaucoma of pseudophakic patients and in pigmentary glaucoma. There is no experience of Bexiprost in inflammatory and neovascular glaucoma or inflammatory ocular conditions. Bexiprost has no or little effect on the pupil, but there is no experience in acute attacks of closed angle glaucoma. Therefore, it is recommended that Bexiprost should be used with caution in these conditions until more experience is obtained.

Bexiprost should be used with caution in patients during the peri-operative period of cataract surgery.

Bexiprost should be used with caution in patients with a history of herpetic keratitis, and should be avoided in cases of active herpes simplex keratitis and in patients with a history of recurrent herpetic keratitis specifically associated with prostaglandin analogues.

Reports of macular oedema have occurred (see section Side effects) mainly in aphakic patients, in pseudophakic patients with torn posterior lens capsule or anterior chamber lenses, or in patients with known risk factors for cystoid macular oedema (such as diabetic retinopathy and retinal vein occlusion). Bexiprost should be used with caution in aphakic patients, in pseudophakic patients with torn posterior lens capsule or anterior chamber lenses, or in patients with known risk factors for cystoid macular oedema.

In patients with known predisposing risk factors for iritis/uveitis, Bexiprost can be used with caution.

There is limited experience from patients with asthma, but some cases of exacerbation of asthma and/or dyspnoea were reported. Asthmatic patients should therefore be treated with caution until there is sufficient experience, see also section Side effects.

Periorbital skin discolouration has been observed, the majority of reports being in Japanese patients. Experience to date shows that periorbital skin discolouration is not permanent and in some cases has reversed while continuing treatment with Bexiprost.

Latanoprost may gradually change eyelashes and vellus hair in the treated eye and surrounding areas; these changes include increased length, thickness, pigmentation, number of lashes or hairs and misdirected growth of eyelashes. Eyelash changes are reversible upon discontinuation of treatment.

Bexiprost contains benzalkonium chloride, which is commonly used as a preservative in ophthalmic products. Benzalkonium chloride has been reported to cause punctate keratopathy and/or toxic ulcerative keratopathy, may cause eye irritation and is known to discolour soft contact lenses. Close monitoring is required with frequent or prolonged use of Bexiprost in dry eye patients, or in conditions where the cornea is compromised. Contact lenses may absorb benzalkonium chloride and these should be removed before applying Bexiprost but may be reinserted after 15 minutes (see section recommended dose).

Paediatric population

Efficacy and safety data in the age group < 1 year (4 patients) are very limited. No data are available for preterm infants (less than 36 weeks gestational age).

In children from 0 to < 3 years old that mainly suffer from PCG (primary congenital glaucoma), surgery (e.g. trabeculotomy/goniotomy) remains the first line treatment.

Long-term safety in children has not yet been established.

Effects on ability to drive and use machines

Instillation of eye drops may cause transient blurring of vision. Until this has resolved, patients should not drive or use machines.

Interaction with other medicaments

Definitive drug interaction data are not available.

There have been reports of paradoxical elevations in intraocular pressure following the concomitant ophthalmic administration of two prostaglandin analogues. Therefore, the use of two or more prostaglandins, prostaglandin analogues or prostaglandin derivatives is not recommended.

Statement on Usage During Pregnancy and Lactation

Pregnancy

The safety of this medicinal product for use in human pregnancy has not been established. It has potential hazardous pharmacological effects with respect to the course of pregnancy, to the unborn or the neonate. Therefore, Bexiprost should not be used during pregnancy.

Breast-feeding

Latanoprost and its metabolites may pass into breast milk and Bexiprost should therefore not be used in breast-feeding women or breast feeding should be stopped.

Side Effects

Summary of the safety profile

The majority of adverse reactions relate to the ocular system, examples iris pigmentation. Other ocular adverse reactions are generally transient and occur on dose administration.

Tabulated list of adverse reactions

Adverse reactions are categorized by frequency as follows: very common, common, uncommon, rare, very rare and not known.

System Organ Class	Very Common	Common	Uncommon	Rare	Very Rare <
Infections and infestations				Herpetic keratitis	
Nervous system disorders			Headache; dizziness		
Eye disorders	Iris hyperpigmentation; mild to moderate conjunctival hyperaemia; eye irritation (burning, grittiness, itching, stinging and foreign body sensation); eyelash and vellus hair changes of the eyelid (increased length, thickness, pigmentation and number of eyelashes)	Punctate keratitis, mostly without symptoms; blepharitis; eye pain; photophobia; conjunctivitis	Eyelid oedema; dry eye; keratitis; vision blurred; macular oedema including cystoid macular oedema; uveitis	Iritis; corneal oedema; corneal erosion; periorbital oedema; trichiasis; distichiasis; iris cyst; localised skin reaction on the eyelids; darkening of the palpebral skin of the eyelids; pseudopemphigoid of ocular conjunctiva	Periorbital and lid changes resulting in deepening of the eyelid sulcus
Cardiac disorders			Angina; palpitations		Angina unstable
Respiratory, thoracic and mediastinal disorders			Asthma; dyspnoea	Asthma exacerbation	
Skin and subcutaneous tissue disorders			Rash	Pruritus	
Musculoskeletal and connective tissue disorders			Myalgia; arthralgia		
General disorders and administration site conditions			Chest pain		

Cases of corneal calcification have been reported very rarely in association with the use of phosphate containing eye drops in some patients with significantly damaged corneas.

Paediatric population

In paediatric patients the safety profile was similar to that in adults. Adverse events seen more frequently in the paediatric population as compared to adults are: nasopharyngitis and pyrexia.

Symptoms and treatment of overdose

Overdose data are limited. The risk of toxicity from inadvertent ingestion is low because the amount contained in the bottles of ophthalmic solution is small. There are no reports of ingestion or ocular administration of large doses of latanoprost in humans.

Symptoms

Apart from ocular irritation and conjunctival hyperemia, no other ocular adverse effects are known if latanoprost is overdosed.

Treatment

If overdosage with latanoprost occurs, treatment should be symptomatic.

Storage Condition

Before opening the cap, keep the bottle in its box in a refrigerator (2°-8° C) protected from light. After opening, keep the bottle in its box in a cool place below 25° C. The contents should be used within 28 days after the dropper is opened. Keep out of reach of children.

Dosage form and packaging available

Bexiprost Eye Drops: Plastic dropper bottle containing 2.5 ml sterile solution. Each ml contains Latanoprost 0.05 mg.

Manufactured by

Beximco pharmaceuticals ltd.

126, Kathaldia, Auchpara, Tongi 1711, Gazipur, BANGLADESH

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